

Tabletop Traffic Junction

Observe, decide, actuate—without conflicting greens

The signal heads reveal the state machine; the display reveals its time.

Lesson	012 — integration project	Time	180–240 minutes
Prerequisites	Lessons 001–011	Board	Arduino Mega 2560
Result	two signal heads, pedestrian pair, count-down, test points	Status	host verified; physical acceptance open
Safety	E1 tabletop model; USB-powered current-limited LEDs only		

Safety checkpoint. This model is not a road controller and must never direct real traffic. Disconnect USB before wiring. Every LED and display segment needs its own resistor. Use only the Mega’s 5 V rail. Stop for heat, odor, resets, unexpected simultaneous greens, or disagreement between the schematic and breadboard. Remove USB power to stop the experiment.

1 Mission and evidence chain

Build two vehicle signal heads, a pedestrian request and signal pair, and a one-digit countdown. The project composes:

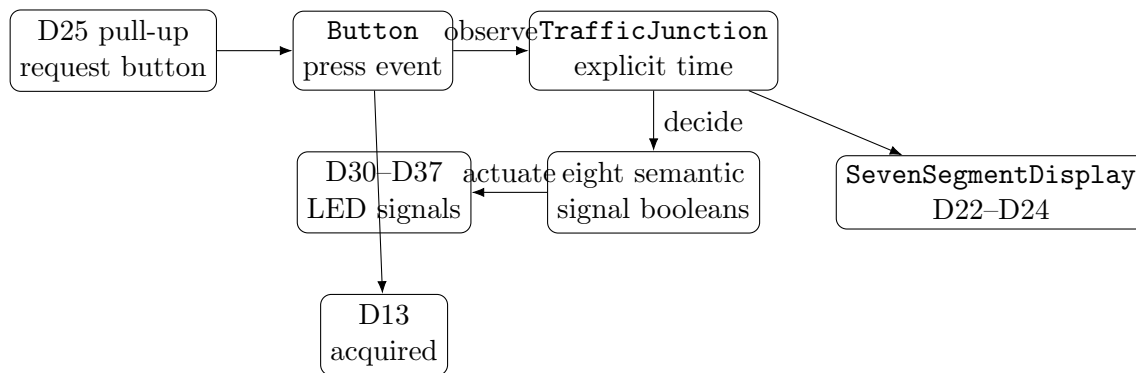
button → TrafficInput → TrafficJunction → TrafficSignals → current-limited LEDs
 nextDeadline → SevenSegmentDisplay → ShiftRegisterOutput → visible digit.

The D13 board LED pulses for 250 ms only after every first-class object initializes. Initialization is transactional: a failed step shuts down every earlier object in reverse dependency order. The heads are the primary diagnostic channel. All-red plus pedestrian stop is the requested running safe state. The countdown is independent supporting evidence: it shows the remaining whole seconds until the next transition. Serial is not required. Named electrical test points are D30 (main red), D32 (main green), D33 (side red), and D35 (side green), each measured relative to GND.

By the end, you can:

- explain why a hardware-neutral state engine receives explicit time;
- trace a latched pedestrian request until it is served;
- prove from observations that both greens are never active together;
- explain why actuation turns permissive signals off before changing red;
- relate a semantic glyph to an atomic 74HC595 latch update;
- distinguish successful acquisition, a commanded all-red state, and a measured electrical all-red state.

2 System composition



The canonical sketch follows the same narrative. `loop()` observes one button event, asks the engine to decide, then actuates the returned snapshot. It never invents signal combinations in Arduino-facing code.

2.1 The state promise

Phase	Vehicle indication	Pedestrian indication
AllRed	both red	stop
MainGreen/MainYellow	named main aspect; side red	stop
SideGreen/SideYellow	named side aspect; main red	stop
PedestrianWalk	both red	walk
PedestrianClearance	both red	stop
Fault	both red	stop

A request is an event, not a held electrical level. The engine latches an accepted request and serves it after the side phase. Each observation also checks that every owned endpoint remains initialized. That check detects lost software ownership; it is not electrical feedback from an LED. A reported circuit-health failure latches **Fault**. The sketch then requests all-red best-effort and performs canonical shutdown. The model runs for two minutes, then releases every owned pin; reset starts a fresh run. No call blocks while a traffic phase elapses.

3 Wiring drawing

4 Parts, limits, and literal wiring

Quantity	Part	Requirement
1	Mega 2560	USB powered
1	74HC595	identified pinout; 0.1 μ F local decoupling
1	one-digit display	common cathode, identified segment pinout
8	signal LEDs	two red/yellow/green heads plus red/green pedestrian pair
16	resistors	eight 330 Ω signal; eight 1 k Ω display
1	momentary button	normally open
1	multimeter	unpowered continuity and powered DC voltage

Mega and signal connections.

LESSON 012 — FOUR WIRING PLATES

Physical locators plus two reference worksheets. Match silkscreen and coordinates; disconnect USB while changing wires.

1. Mega 2560 physical header locator

USB

D13 header / built-in LED

D22-D53 PAIRED HEADER

inner outer

○ 22 DS ○ 23 SHCP
○ 24 STCP ○ 25 BUTTON
○ 26 ○ 27
○ 28 ○ 29
○ 30 main ○ 31 main Y
○ 32 main ○ 33 side R
○ 34 side ○ 35 side G
○ 36 WALK ○ 37 STOP
38 ... 52 39 ... 53

MEGA 2560

POWER header: 5V and GND

POWER JACK

USB and barrel jack are at left; use the board's printed socket labels.

2. 74HC595 across the breadboard trench

+5V RAIL

GND RAIL

row 20: 1 QB
row 21: 2 QC
row 22: 3 QD

16 VCC -> 5V
15 QA
14 DS from D22

TRENCH

row 23: 4 QE
row 24: 5 QF
row 25: 6 QG
row 26: 7 QH
row 27: 8 GND

13 OE -> GND
12 STCP from D24
11 SHCP from D23
10 MR -> 5V
9 QH' unused

Notch up. One 0.1 uF capacitor directly from pin 16 to pin 8.

3. Signal breadboard — physical top view

A-E connected per row; F-J connected per row; trench separates them.

+ RAIL

GND RAIL

10-12 D30 main R	A10--330R--A11; C11 LED C12; A12--GND
14-16 D31 main Y	A14--330R--A15; C15 LED C16; A16--GND
18-20 D32 main G	A18--330R--A19; C19 LED C20; A20--GND
24-26 D33 side R	A24--330R--A25; C25 LED C26; A26--GND
28-30 D34 side Y	A28--330R--A29; C29 LED C30; A30--GND
32-34 D35 side G	A32--330R--A33; C33 LED C34; A34--GND
38-40 D36 WALK	A38--330R--A39; C39 LED C40; A40--GND
42-44 D37 STOP	A42--330R--A43; C43 LED C44; A44--GND
48/50 D25 BUTTON	button straddles trench: E48 to D25; F50 to GND

A B C D E | F G H I J

LED long leg in the lower-numbered signal row; bridge split rails like-to-like.

4. Display mapping — specimen gate

Proceed only after the marking confirms common cathode.

QA -> 1k -> a -> pin ____

QB -> 1k -> b -> pin ____

QC -> 1k -> c -> pin ____

QD -> 1k -> d -> pin ____

QE -> 1k -> e -> pin ____

QF -> 1k -> f -> pin ____

QG -> 1k -> g -> pin ____

QH -> 1k -> DP -> pin ____

common cathode pin(s) ____ -> GND

Eight 1k resistors; never borrow another display's pinout.

Figure 1: Four complementary pencil plates: physical Mega header locator, 74HC595 placement worksheet, physical signal breadboard coordinates, and specimen-gated display mapping. Match the selected display's identified package pins to the named a-g/DP nets before inserting it.

Function	Mega pin	Literal path
request	D25	D25 to button to GND; internal pull-up
acquisition	D13	built-in LED; 250 ms pulse after every object initializes
main head	D30/D31/D32	pin to 330 Ω to red/yellow/green anode; every cathode to GND
side head	D33/D34/D35	same order and resistor rule
pedestrian	D36/D37	walk green/stop red, each through 330 Ω

74HC595 and display connections.

Function	Connection	Required interpretation
serial data/clock/latch control	D22 to DS, D23 to SHCP, D24 to STCP MR to 5 V; OE to GND	<code>ShiftRegisterPins{22,23,24}</code> never leave either floating; see startup limit below
power segments	VCC to 5 V; GND to GND; 0.1 μ F across them Q0-Q6 to a-g, Q7 to decimal point	place capacitor at the IC each path has its own 1 k Ω resistor
display return	both common cathodes to GND	verify the exact package pinout

The supplied one-digit display family has multiple package mappings. Read its marking and datasheet, verify

that it is common-cathode, and write its actual pin numbers beside the named a-g/DP nets in the drawing before insertion. If those facts are unknown, stop before powering the display; the vehicle and pedestrian wins remain usable without it. With a conservative 3 mA per lit display segment and 10 mA per signal LED, the worst expected visible state is under 55 mA; the 330 Ω and 1 k Ω paths keep this USB-powered model comfortably bounded.

Safety checkpoint. With OE tied low, the 74HC595 storage register can present an unspecified pattern before the first software latch. The three-wire lesson circuit therefore cannot promise a blank display at power-up or reset. Its atomic-update claim begins only after initialization. A later hardened adapter must own OE and hold it disabled with a pull-up until a known blank value has been latched.

5 Three quick wins

Win 1 — make the junction breathe. Wire the two vehicle heads and run the sketch. Both reds begin together, then permission moves through green, yellow, and all-red without ever showing two greens. If one color or direction is wrong, fix that literal D30–D35 path before adding anything else. *Expected sight:* main green with side red, then main yellow, both red, side green with main red, side yellow, and both red again.

Win 2 — let a person ask to cross. Add D36/D37 and the D25 button. Press and release once during main green. The later walk light is the payoff: it appears only if the button event, state engine, and both pedestrian outputs are working together. *Expected sight:* STOP stays on through the vehicle sequence; after both reds return, STOP yields to WALK, then returns.

Win 3 — reveal the clock. Identify the exact common-cathode display, then add the 74HC595 and one 1 k Ω resistor per segment. The digit now counts down to the same transitions already visible in the heads. A wrong segment is local to its named Q output, resistor, and display pin. *Expected sight:* one stable digit at a time, decreasing toward each lamp change without a mixed intermediate glyph.

After two minutes every lamp and the display go dark as the sketch releases its pins. Press reset and look for the short D13 acquisition pulse to play again. These wins are the learner path; deterministic traces and failure injection remain automated repository checks.

6 Build and run from the CLI

```
make arduino-Lesson012TrafficJunction
make upload-Lesson012TrafficJunction PORT=/dev/ttyACM0
make monitor-describe PORT=/dev/ttyACM0
make serial-log PORT=/dev/ttyACM0 \
  SERIAL_LOG=build/evidence/lesson012-optional.log
```

Serial capture is optional context. The required proof is the visible and electrical evidence below. Build and upload targets must remain the canonical path; no IDE-specific step is part of the lesson.

6.1 The narrative run loop

Listing 1: Canonical observe–decide–actuate flow.

```
adk::SevenSegmentGlyph countdownGlyph (uint8_t seconds);
bool          circuitHealthy  ();
bool          requestAllRed   ();
void          shutdownJunction ();
```

```

    void                haltJunction    ();
} // namespace

void setup ()
{
    halted = !acquireJunction ();

    if (!halted)
    {
        runStartedAt = adk::TimePoint (millis ());
    }
}

void loop ()
{
    if (halted)
    {
        return;
    }
}

```

The engine's **TrafficSignals** is the sole actuator model. The **showSignals()** adapter first removes green, walk, and yellow permission; then it establishes all-red; only then does it present the requested state. This ordering reduces transition hazards when writes succeed. It cannot guarantee an electrical state after a driver, wire, LED, or supply failure. The fault path attempts every all-red write even if an earlier write fails, blanks the display, and releases every endpoint to high impedance.

7 Watch the story unfold

Predict	Observe	Interpret
Acquisition completes	D13 pulses for 250 ms	every project object initialized; this does not prove the signal pin levels
Startup is all-red	both red heads and pedestrian stop illuminate before any green	the first semantic state was actuated; measure test points before calling it electrical proof
Only one vehicle direction proceeds	watch an entire main/side cycle	green moves from one direction to the other only through yellow and all-red
One request is remembered	press and release D25 once during main green; wait without holding it	a later pedestrian walk proves event capture and latching
The display changes atomically	watch the countdown at a phase boundary	mixed or torn segments indicate wiring, resistor, or latch trouble
The show has a finale	let the two-minute run finish	every owned output is released and every light goes dark; reset starts another show

The lamps explain the code while it runs: the D13 pulse introduces the show, the two heads reveal legal traffic motion, the pedestrian pair reveals a remembered request, and the digit reveals when the next change will happen.

8 Try these variations

Once all three wins work, predict what the model will do before each variation, then watch the lamps and digit:

1. Press once during main green and guess when walk will appear.
2. Press several times during one phase; the crossing remains one orderly request rather than skipping ahead.
3. Reset during green. The next visible traffic state is all-red, preceded by the short D13 pulse.
4. Let the two-minute show finish. Everything goes dark; reset is the only way to begin another run.

9 Diagnosis

Observation	Likely layer	Next bounded check
No LEDs and blank display	power or acquisition	remove USB; inspect rails, polarity, pin map, and resource conflicts
Heads correct, digit wrong	display encoding/wiring	run lesson 010; compare Q0–Q7 with a–g/decimal point
Digit tears during change	latch wiring	inspect STCP D24 and shared ground; do not add delays to hide it
Request never served	button/event path	observe raw D25 and repeat lesson 003 debounce evidence
Both greens visible	actuator wiring or severe defect	remove USB immediately; unpowered continuity from D32/D35 to the intended LEDs
All-red forever	startup/fault/configuration	reproduce the same timestamps in host tests and inspect status
Unexpected resets	current or wiring	remove USB; measure resistors and inspect for rail shorts before reapplying power

10 Make it yours

1. Make cardboard signal-head masks so each LED shines through its own round opening.
2. Change only the configured phase durations and invent a fast “tiny town” cycle or a leisurely “parade day” cycle.
3. Show a dash during all-red and digits during moving phases.
4. Add a second cardboard pedestrian button cap labelled *WAIT*; keep both caps on the same safe D25 switch rather than adding wiring.

11 What to carry forward

The reusable idea is not “a traffic light.” It is the boundary between a deterministic decision and a transactional physical presentation. The engine speaks in domain signals; endpoint-owning adapters perform safe ordering; the circuit supplies evidence alongside its useful behavior. The same separation supports later environmental, access, rover, greenhouse, telemetry, and inert cue projects.

Primary source paths: Traffic junction API, seven-segment API, shift-register API, and canonical example.